

A Comparative Study of Standard spaCy and spaCy-Transformers for Modern NLP Tasks

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1 WORK TO DO

The goal of this project is to assess the trade-offs between spaCy’s standard statistical models for NLP and the spaCy-Transformers extension. The spaCy-Transformers extension incorporates Hugging Face transformer models into spaCy. While transformer models offer state-of-the-art accuracy across a variety of tasks at the cost of higher computational requirements, standard spaCy models are renowned for their speed and efficiency in production-ready NLP applications.

Our study will focus on three representative tasks in NLP:

- Named Entity Recognition (NER): Detecting people, organizations, and locations in text.
- Text Classification: Assigning categories such as sentiment or topic labels to documents.
- Semantic Similarity: Comparing sentence and document embeddings for search and retrieval.

Our methodology involves combining knowledge from prior research with results from targeted experiments we conduct. To showcase these outcomes, we will prepare demo for each task. The steps are as follows:

- Select datasets applicable to each task. For example, a news dataset for NER, a review dataset for classification, and a sentence-pair dataset for similarity.
- Run both spaCy standard pipeline and spaCy pipeline with transformers on the same data.
- Measure results in accuracy-related quantities like F1 score, classification accuracy, and similarity scores, as well as practical considerations like how fast the models calculate, how much memory they use, and how large the model files are.
- Compare outputs to determine where the transformer models significantly add value and where the cost in speed and resources potentially is not warranted.

We will be uncovering the accuracy-speed balance between the basic spaCy models with the transformer-based ones. More specifically, we are interested in uncovering how much accuracy is improved through the use of transformers and how more time and computer resources are being utilized. The exploration shall provide practical recommendations on when it is worthwhile to use the fast and light-weight basic spaCy models, such as in chatbots or time-critical systems, and on what occasions it is worthwhile to choose the slow but accurate transformer models, e.g., in research work, in which accuracy is more important as compared to speed.

2 WORK COMPLEXITY

We expect it to take around 25–30 hours in total for the project. Around a week will be spent on background reading and data preparation, a second and third on writing demos and conducting experiments, and the fourth on results interpretation and writing up the report. The workload is roughly to the duration of an entire semester and is expected to yield both useful, hands-on experience and significant, real-world outcomes.